MODEL LAB · 2026-10-16

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)
served prior

11
priors compared

26,271
held-out plate appearances

3.07%
held-out HR rate

0.6059
best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-04, scored on the 26,271 that came after (actual HR rate 3.07%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5794	1.65×	0.02968	0.13619	0.00571
KNN comps Prior is the pooled rate of the nearest comparables.	0.6059	1.56×	0.02962	0.13488	0.00390
SVR Support-vector regression on the same feature matrix.	0.5972	1.56×	0.02967	0.13558	0.00596
Random forest Forest regression; a learned similarity metric.	0.5982	1.55×	0.02964	0.13516	0.00523
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6059	1.56×	0.02962	0.13488	0.00390
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6059	1.56×	0.02962	0.13488	0.00390
Linear regression Ordinary least squares on the same features.	0.6017	1.58×	0.02965	0.13518	0.00398
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6027	1.65×	0.02966	0.13524	0.00623
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6007	1.60×	0.02964	0.13512	0.00470
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5988	1.58×	0.02966	0.13527	0.00582

BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

SHRINKAGE K PER PRIOR

league **251** knn **251** svr **251** rf **251** core **251** ensemble **251** linear **251** logistic **251** nn **251** xgb **251**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (2,786 PA, actual HR rate 2.37%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5860	1.36×	0.02314	0.11229	0.01037
KNN comps Prior is the pooled rate of the nearest comparables.	0.6232	1.97×	0.02306	0.10992	0.00977
SVR Support-vector regression on the same feature matrix.	0.6017	1.82×	0.02325	0.11321	0.01716
Random forest Forest regression; a learned similarity metric.	0.6106	2.12×	0.02316	0.11068	0.01542
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6232	1.97×	0.02306	0.10992	0.00977
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6232	1.97×	0.02306	0.10992	0.00977
Linear regression Ordinary least squares on the same features.	0.6211	1.97×	0.02326	0.11135	0.01418
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6307	1.66×	0.02325	0.11095	0.01676
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6236	1.82×	0.02308	0.11000	0.01268
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6196	1.82×	0.02326	0.11125	0.01972

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Jonathan Aranda 14.4%, Junior Caminero 14.1%, Jo Adell 13.8%, Patrick Bailey 13.2%, Victor Mesa Jr. 12.8%, Chase DeLauter 12.0%
Flat league	6/6	Jonathan Aranda 13.6%, Junior Caminero 13.5%, Jo Adell 13.1%, Victor Mesa Jr. 12.6%, Patrick Bailey 12.5%, Chase DeLauter 12.1%
KNN comps	6/6	Jonathan Aranda 14.5%, Junior Caminero 14.0%, Jo Adell 13.5%, Patrick Bailey 13.2%, Victor Mesa Jr. 12.9%, Chase DeLauter 12.1%
SVR	6/6	Junior Caminero 15.2%, Jo Adell 14.1%, Jonathan Aranda 14.0%, Victor Mesa Jr. 13.5%, Patrick Bailey 13.0%, Chase DeLauter 11.9%
Random forest	6/6	Junior Caminero 14.4%, Jonathan Aranda 14.3%, Jo Adell 13.7%, Victor Mesa Jr. 13.4%, Patrick Bailey 12.7%, Chase DeLauter 12.0%
Core (KNN+SVR)	6/6	Jonathan Aranda 14.5%, Junior Caminero 14.0%, Jo Adell 13.5%, Patrick Bailey 13.2%, Victor Mesa Jr. 12.9%, Chase DeLauter 12.1%
Forest blend	6/6	Jonathan Aranda 14.5%, Junior Caminero 14.0%, Jo Adell 13.5%, Patrick Bailey 13.2%, Victor Mesa Jr. 12.9%, Chase DeLauter 12.1%
Linear regression	6/6	Jonathan Aranda 14.3%, Junior Caminero 14.1%, Jo Adell 13.8%, Victor Mesa Jr. 13.2%, Patrick Bailey 13.0%, Chase DeLauter 11.9%
Logistic regression	6/6	Junior Caminero 14.5%, Jo Adell 14.2%, Jonathan Aranda 14.1%, Victor Mesa Jr. 13.1%, Patrick Bailey 12.6%, Chase DeLauter 11.8%
Neural net (torch)	6/6	Junior Caminero 14.5%, Jonathan Aranda 14.2%, Jo Adell 14.0%, Victor Mesa Jr. 13.3%, Patrick Bailey 12.8%, Chase DeLauter 11.8%
XGBoost	6/6	Junior Caminero 15.0%, Jonathan Aranda 14.3%, Jo Adell 14.0%, Victor Mesa Jr. 13.1%, Patrick Bailey 12.6%, Chase DeLauter 11.8%

Hitter by hitter

The top 40 bats by the served probability.

HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1 Jonathan Aranda TB slate	**14.4%**	13.6%	14.5%	14.0%	14.3%	14.5%	14.5%	14.3%	14.1%	
2 Junior Caminero TB	**14.1%**	13.5%	14.0%	15.2%	14.4%	14.0%	14.0%	14.1%	14.5%	
3 Jo Adell CLE slate	**13.8%**	13.1%	13.5%	14.1%	13.7%	13.5%	13.5%	13.8%	14.2%	
4 Patrick Bailey CLE	**13.2%**	12.5%	13.2%	13.0%	12.7%	13.2%	13.2%	13.0%	12.6%	
5 Victor Mesa Jr. TB	**12.8%**	12.6%	12.9%	13.5%	13.4%	12.9%	12.9%	13.2%	13.1%	
6 Chase DeLauter CLE	**12.0%**	12.1%	12.1%	11.9%	12.0%	12.1%	12.1%	11.9%	11.8%	
7 Travis Bazzana CLE	**11.6%**	11.6%	11.5%	11.2%	11.4%	11.5%	11.5%	11.4%	11.1%	
8 Nathaniel Lowe CLE	**11.5%**	11.3%	11.6%	11.4%	11.5%	11.6%	11.6%	11.5%	11.4%	
9 Yandy Díaz TB	**11.0%**	11.3%	11.1%	11.5%	11.2%	11.1%	11.1%	11.2%	11.0%	
10 José Ramírez CLE	**10.4%**	10.7%	10.5%	10.6%	10.6%	10.5%	10.5%	10.8%	11.1%	
11 Petey Halpin CLE	**9.6%**	10.5%	9.6%	10.0%	9.8%	9.6%	9.6%	10.7%	10.2%	
12 Liam Hicks TB	**7.6%**	8.6%	7.5%	8.4%	7.5%	7.5%	7.5%	8.0%	7.8%	
13 Brayan Rocchio CLE	**7.4%**	7.9%	7.3%	7.3%	6.9%	7.3%	7.3%	7.1%	7.1%	
14 Jonny DeLuca TB	**6.6%**	7.1%	6.4%	6.5%	6.2%	6.4%	6.4%	6.5%	6.7%	
15 Richie Palacios TB	**6.4%**	7.3%	6.5%	6.2%	5.9%	6.5%	6.5%	6.3%	6.3%	
16 Steven Kwan CLE	**5.3%**	8.6%	5.6%	3.7%	4.3%	5.6%	5.6%	3.7%	5.3%	
17 Taylor Walls TB	**4.4%**	6.0%	4.4%	3.7%	3.9%	4.4%	4.4%	4.6%	4.5%	
18 Chandler Simpson TB	**2.6%**	5.8%	3.4%	1.9%	1.5%	3.4%	3.4%	1.5%	3.0%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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